

Gennoor Tech

TRAIN. INNOVATE. BUILD.

GENNOOR ACADEMY · SELF-PACED ·
36 MIN

AI in Manufacturing

A ~36-minute playbook for plant managers and VPs of manufacturing. 6 plays, 3 anti-plays, the OT/IT bridge filter, and 4 trust trip-wires. Voice: Emma.

8

CHAPTERS

36 min

DURATION

Intermediate

LEVEL

FREE

CERTIFICATE

Course Overview

For plant managers, VPs of manufacturing, plant engineers, quality + maintenance leaders, and operations directors. Manufacturing AI is plant-floor reality, not vendor demo — sensors fail, lighting changes, operators make judgement calls, networks go down. Six plays that consistently ship: predictive maintenance (with sensor coverage decisions + the narrative layer engineers actually act on), quality vision AI (with the operator labelling loop + false-confidence year-2 discipline), process optimisation (AI surfaces, operators decide, with 3 overfitting guards), plant-floor supply chain integration (3 signal flows + 3 re-planning cadences), safety AI as support-not-surveillance (with the charter signed by safety + ops + union before deployment), pragmatic digital twin (6-month bounded scope, ONE specific use case, plant engineers in design).

FORMAT: Self-paced · 8 chapters · audio-narrated slides · interactive end-of-course builder · free certificate on completion

Who Is This For?

1

Plant managers · VPs
Manufacturing

2

Operations leadership

3

Process + quality
engineers

4

IT/OT leaders

Learning Outcomes

- 1 6 plant-floor plays that ship — predictive maintenance, vision quality, energy, scheduling, safety, throughput
- 2 3 pitches that disappoint — and the OT/IT bridge filter that separates them
- 3 How to design the OT/IT bridge that makes AI on the shop floor possible
- 4 The digital-twin reality applied honestly — what works, what's marketing, what breaks at production scale
- 5 The ISA-95 stack as the deployment frame — where AI plugs in safely vs where it threatens uptime
- 6 A 12-month plant roadmap with OEE and quality KPIs as the success measure
- 7 The trust trip-wires — worker safety, equipment integrity, regulatory disclosure

8-Chapter Curriculum

Self-paced, audio-narrated slide format. Each chapter ends with a "Monday move" the learner can apply that week.

01

The manufacturing AI landscape

CHAPTER 1 · 5 MIN · NARRATED + INTERACTIVE

- 6 plays
- 3 anti-plays
- OT/IT bridge filter (plant engineering + IT + AI team in room from day one).

02

Predictive maintenance for plants

CHAPTER 2 · 5 MIN · NARRATED + INTERACTIVE

- 3 sensor coverage decisions
- the narrative layer engineers act on
- 3 project killers (data quality
- alert fatigue
- CMMS disconnect).

03

Quality vision AI

CHAPTER 3 · 5 MIN · NARRATED + INTERACTIVE

- 3 messy challenges (lighting
- drift
- novel defects)
- 3-stage operator labelling loop
- false-confidence year-2 failure mode + 3 disciplines.

04

Process optimisation with AI

CHAPTER 4 · 5 MIN · NARRATED + INTERACTIVE

- 3 signals AI surfaces (multi-parameter
- drift
- best-shift)
- operator-led experiment design
- 3 overfitting guards.

05

Plant-floor supply chain integration

CHAPTER 5 · 5 MIN · NARRATED + INTERACTIVE

- 3 signal flows (demand
- material
- equipment)
- 3 re-planning cadences (strategic monthly
- tactical weekly
- operational real-time)

06

Safety AI

CHAPTER 6 · 5 MIN · NARRATED + INTERACTIVE

- 3 use cases (PPE
- near-miss
- ergonomic)
- the line nobody crosses (support, not surveillance)
- 3 failure modes specific to safety AI.

07

Digital twin: pragmatic, not megaproject

CHAPTER 7 · 5 MIN · NARRATED + INTERACTIVE

- The megaproject trap
- 3 targeted twins that ship in 4-6 months
- 3 discipline rules (6-mo TTV
- 1 bounded use case
- plant engineers in design).

08

Making it stick: your plant AI roadmap

CHAPTER 8 · 6 MIN · NARRATED + INTERACTIVE

- 12-month plant rollout
- 4 trust trip-wires
- interactive Markdown builder for your plant manager + VP Manufacturing.